Electrical particle generator

Abstract

ELECTRICAL PARTICLE GENERATOR

ABSTRACT
An electrical particle generator comprising a non-magnetic pipe in a closed loop having a substantial amount of magnetized particles encapsulated therein. A magnetic accelerator assembly positioned on said pipe having an inductive primary winding and a low voltage input to said winding. A secondary winding positioned on said pipe opposite to said primary winding. Upon voltage being applied to said primary winding said magnetized particles are passed through said magnetic accelerator assembly with increased velocity. The velocity accelerated particles continuing through said pipe induce an electrical voltage/current potential as they pass through the secondary winding. The increased secondary voltage is utilized in an amplifier arrange-ment.

CA1213671A
Canada

Find Prior Art Similar

Other languages: French

Inventor: Stanley A. Meyer

Current Assignee : Individual

Worldwide applications

1982 EP 1983 ~~CA~~ JP

Application CA000420909A events

1983-02-04	Application filed by Individual
1986-11-04	Application granted
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Status	Expired

Info: Patent citations (5), Cited by (2), Legal events, Similar documents, Priority and Related Applications

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Classifications

H01F30/10 Single-phase transformers

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Claims (16)

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CLAIMS

1. An electrical particle generator comprising;
a non-magnetic closed-loop tubing, passing magnetic lines of force, a particle accelerator positioned adjacent to one end of said tubing, a secondary inductive winding positioned adjacent to said tubing opposite to the position of said accelerator, a substantial volume of magnetized elements encapsulated in said tubing;
said accelerator having means to provide a low input voltage; and said secondary winding having means for tapping a high voltage/current for utilization.

2. The electrical particle generator of Claim 1 wherein said magnetized elements are particles suspended in a fluid medium.

3. The electrical particle generator of Claim 2 wherein said magnetized elements is a gas.

4. The electrical particle generator of Claim 2 wherein said magnetized elements is a liquid.

5. The electrical particle generator of Claim 2 wherein said magnetized elements is a solid.

6. The electrical particle accelerator of Claim 1 wherein said particle accelerator is an electrical magnetic particle accelerator.

7. The electrical particle generator of Claim 1 wherein said particle accelerator further comprises a primary inductive winding, and a low voltage input to said winding.

8. The electrical particle generator of Claim 4 wherein said low voltage input is a variable voltage input for controlling the magnitude of the voltage/current induced in said secondary winding.

9. The electrical particle generator of Claim 4 wherein said input voltage to said primary winding is direct current, and wherein said output voltage is a direct current/voltage.

10. The electrical particle generator of Claim 4 wherein said input voltage to said primary winding is alternating current and wherein said output current/voltage is an alternating current/
voltage.

11. The electrical particle generator of Claim 4 wherein said input voltage to said primary winding is pulsed, and wherein said output current/voltage is a pulsed current/voltage.

12. The electrical particle generator of Claim 4 wherein said input voltage to said primary winding has variable waveform, and wherein said output current/voltage will have a similar waveform as said input voltage waveform.

13. The electrical particle generator of Claim 4 wherein said particle accelerator further comprises a magnetic forming core, and wherein said core has an opening slightly greater than the outside diameter of said tubing and adapted to receive the same.

14. The electrical particle generator of Claim 1 said tubing has at said opposite end three parallel discrete branches and wherein said secondary coil is three discrete coils with three discrete outputs one of each positioned over one of each of said tubing branches thereby providing a three phase output.

15. The electrical generator of Claim 1 wherein said particle accelerator comprises mechanical pump means, and wherein said magnetized elements are permanently magnetized elements.

16. An assembly for providing a magnetized particle for utilization in an electrical particle generator comprising a chamber and a pair of magnetizable material electrodes positioned therein, a source of voltage/current of opposite polarity applied to said electrodes, said magnetizable material becoming vaporized upon the application of said voltage/current in said electrodes, a pipe connected to said chamber, and means directing said vaporized particles to said pipe, a magnetic field generator having the other end of said pipe positioned within its magnetic field, and wherein said vaporized particles entering and being expelled through said pipe become magnetized by passing through said magnetic field generator.

Description

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ELECTRICAL PARTICLE GENERATOR

1 ~ BACKGROUN~ AND PRIOR ART

2 The prior art teachings expound the fundamental principle

3 that a magnetic field passing through inductive windings will ~ ~enerate a voltage/current or enhance the voltage thereacross if the.winding is a secondary winding.
6 It is also taught by the prior art that a magnetic element 7 in a primary inductive f~eld will be attracted at one end of the coil and repelled at the other. That is a moving magnetic 9 element will be accelerated in motion by the attraction and repulsion of the magnetic field of the primary inductive winding.
11 In t convention l step-up transfer the voltage across ~ .

lZ13671 ;, l ~ the secondary is a function of the nu~ber of turns of the 2 ¦ secondary relative to the number of primary turns. Other 3 ¦ factors are the size of the turns and whether the core is air

4 ¦ or magnetic.

5,1 SUMMARY OF INVENTI~N

6 ¦ The present i~vention utilizes the basic principle of a

7 particle accelerator and the principle of inducing a voltage in

8 a secondary winding by passing a magnetic element therethrough.

9 The structure comprises a primary voltage inductive winding having a magnetic forming core and a low voltage input. A
11 secondary winding comprises a greater number of turns than the 12 primary and has an output for utilization of the voltage induced 13 therein.
14 The primary inductive winding and core are positioned around a first position on an endless - closed loop, non-magnetic 16 pipe. The secondary windings are positioned around an opposite 17 position of said endless pipe.
18 The pipe is filled with discrete magnetic elements having l9 a magnetic polarized charge placed thereon. The magnetic elements are particles and preferably gas.
21 The magnetic elements due to their polarization will sustain 22 some motion. As the particles approach the accelerator assembly, 23 the primary coil, the field therein attracts the particles and 24 accelerate the particles to enter the gap in the coil. As the magnetic element proceeds, the repulsion end of the accelerator ~ lZ136';'1 1 will impart further magnetic force to the particles. This 2 magnetic attraetion and the repulsion considerably enhances 3 the motion of the discrete particles within the pipe. The 4 particles are ejected from the area of the accelerator with 5 an increased velocity motion.
6 The magnetic elements proceed in the closed loop pipe at 7 a speed considerably greater than their normal movement due r to the accelerator action. As the magnetic elements pass through 9 the core of the secondary winding there is induced a voltage

10 therein. In this way a much greater voltage is induced in the

11 secondary winding of the transformer.

12 OBJECTS

13 It is principal object of the present invention to provide

14 an electrical generator capable of producing a voltage/current

15 much greater in magnitude hereintofore possible.

16 Another object of the present invention is to provide such

17 an electrical generator utilizing magnetized elements and a

18 magnetic acceleratox.
l9 Another object of the present invention is to provide such 20 an electrical generator that can control the amplitude of the 21 output.
22 Anothe~ ob;ect of the present invention is to provide such 23 an electrical generator that may be utilized with direct current, 24 alternating current, p~lsed, or other configurations of waveforms.
25 Another object of the present invention is to provide such . .

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1 an electrical generator that may be utilized in a single phase 2 or three phase electrical system.
3 Still another object of the present invention is to provide a genera~or for developing magnetized particles for use in an electrical particle accelerator.

-6 A further object of the present invention is to provide 7 such an electrical generator ~hat utilizes components readily available are adaptable to a simplified embodiment.
Other objects and features of the present invention will become apparent from the following detailed description where 11 taken in conjunction with the drawings in which:
12 BRIEF DESCRIPTION OF'DRAWINGS
13 Figure 1 is a simplified illustration of the principles 14 of the invention shown partly in crossection and partly pictorially.
16 Figure 2 is an electrical schematic illustration of the 17 embodiment of Figure 1.
18 Figure 3 is an illustration similar to Figure 2 but

19 adaptable to 3 phase.
Figure 4 is a first alternati~e arrangement in a preferred 21 illustration of the invention.
22 Figure~5 is another alternative arrangement in a preferred 23 illustration of the invention.
24 Figure 6 is another alternative arrangement in a preferred illustration of the invention.

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1 Figure 7 is another alternative arrangement in a preferred 2 illustration of the invention.
3 Figure 8 is another alternative arrangement in a preferred 4 illustration of ~he invention.
Figure 9 is an alternative arrangement for a magnetic drive accelerator assembly.
7 Figure 10 is a mechanical illustration of a magnetic particle 8 generator for developing magnetic particles utilized in the 9 present invention.
DETAILED DESCRIPTION OF INVENTION
11 With particular reference to Figures 1 and 2 there is 12 illustrated the in~ention in its mostly simplified schematic 13 embodi~ent.
14 The system of the invention comprises a primary winding coil magnetic accelerator assembly 10, a closed loop non-magnetic 16 pipe 30, and a secondary winding 20.
17 The magnetic accelerator assembly comprises primary windings 18 12 a magnetic core 14 and voltage taps 16. The primary windings 19 are positioned around end 32 of closed loop pipe 30 or tubing of non-magnetic material.
21 At the opposite end 34 of the closed loop pipe 30 there is 22 positioned thereon secondary windings 20. A voltage tap ~2 on 23 secondary winding 20 permits the utiliæation of the voltage 24 induced therein.
Within the pipe 30 ~here is encapsulated a substantial . , . ~ S . . , ~

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1 quantity of magnetic elements 40 of Figure 2. The elements 40 2 must be sufficiently light to be freely mobile. Therefore, the 3 elements may be particles suspended in a fluid medium such as 4 gas, liquid, or light-weight movable solid particles and more preferably gas. In the applicati~n of suspended solid magnetic 6 particles it may be desireable to evacuate the tubing to reduce 7 the resistance to its flow. ~he magnetized elements 40 are , discrete elements, that is, each particle or a miniscule part 9 thereof is separately magnetized. Further, the action as here-inafter described is an action upon each par~icle and not upon 11 the mass.
12 The voltage applied to terminals 16 of the primary winding 13 12 is a low voltage the magnitude of which may be utilized as an 14 input signal control. By varying the input voltage the accelera-tor will vary the velocity of the speed of the particles; which, 16 in turn vary the magnitude of the voltage/current output of the 17 secondary winding 20.
r~ 18 The output 22 of the secondary transformer 20 is a high l9 voltage/current output.
It can be appreciated that the system of Figures 1 and 2 21 utilizing a sin`gle c~osed loop will provide a single phase output 22 in the secondary winding. With particular reference to Figure 3 23 there is illustrated a closed loop system having three parallel 24 non-magnetic tubes 31, 33, and 35. On each tube or pipe there is a respective secondary winding 21, 23, and 25.
26 Each of the secondary windings 21, 23, and 25 are a single 27 phase much as that shown in Figures 1 and 2. The three pipes, 28 having a common iunction at its input and at its output, with 29 the respective three secondary pick-up windings provide a :;' ". . ' .

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1 balance three-phase (3) electrical system.
2 The electrical power generator illustrated in Figure 4 is in 3 electrical operation identical to that in Figures 1 and 2. The 4 physical configuration of the arrangement is adapted to be utiliz-ed in a high moisture environment. An insulating coa~ing 45 com-6 pletely covers the pipe 30, as well as all electrical windings.
7 In Figure 4 there is illustrated that increasing the number r' 8 of coils with a given size winding the voltage/current output is 9 increased. The physical configuration illustrated is such that the vertical aspects as well as the horizontal aspect are utilized.
11 In this way a large diameter pipe can be used with a substantial 12 number of turns of heavy-gauge high-current wire.
13 With particular reference to Figure 5 there is illustrated 14 a coil arrangement 49 that utilizes the entire magneti~ flux in the closed loop tubing 47. The configuration is that of a 16 coaxial arrangement with the primary winding 43 as a center core.
17 Figure 6 illustrates a concentric spiral configuration of 18 the tubing 50 having the seco~dary windings 53 wound over it's 19 entirety.
In Figure 7 the particle accelerator 10 is wound over t~e 21 tubing 30 much in the same manner as the Figures 1 and 2.
22 However, in this embodiment the tubing 30 is a continuous closed 23 loop but in a series, parallel configuration. That is electri-24 cally there are three secondary windings with three outputs : ~ ~whereas the ubing 30 is in series through ~he three secondary `.' . . ' "`

; ~ 213 l windings.
2 In Figure 8 the reverse of that configuration of Figure 7 3 is shown. That is the several pick-up coils are wound in series;
4 whereas the tubing 80 is not a continuous tubing as that of the other configuration. Principally, there is disclosed an input 6 manifold 82 and an output manifold 84 with the several tubings 7 60a xxx n interconnecting the two manifolds. Each one of the f~ aforesaid secondary coils 70a xxx n are wound over its respective 9 tubing.
The magnetic particle accelerator 10 may comprise configura-11 tions and structures distinctive from the electrical accelerator 12 o~ Figure 1. With reference to Figure 9 there is illustrated a 13 mechanical particle accelerator 100. In this embodiment the 14 magnetic particles 102 are permanently magnetized prior to being encapsulated in non-magnetic pipe 110. The particles 102 are 16 accelerated by fan blade or pump 104 rotated by mechanical drive 17 assembly 106. The mechanical drive for the assembly 106 may ~ 18 comprise a pulley 112 attached by a belt or like to an electric l9 motor. A sealing bearing 114 retains the particles 102 within 22 the plpe l .
., ~ ~ . , ... Il,~, lZ13~71 1 It was stated above, the particles traversing the secon~ary 2 coils a voltage/current is generated therein. It must be 3 appreciated that the particles are not in actuality traversing 4 the coils but are in fact traversing the magnetic field of the S coils.
6 Again, the pipe 30 is described as a non-magnetic pipe.
7 There are certain non-magnetic pipes that would not be operable 8 in the present invention; That is, the pipe 30 must be capable 9 of passing magnetic lines of force; and that it is these magnetic lines of force that traverse the inductive field of the secondary 11 windings 20 to induce a voltage/current therein.

12 A significant feature of the invention illustra~ed in the 13 various embodiments hereintofore described, as the generation 14 of the magnetized particles encapsulated in the tubing. With reference to Figure 10, there is illustrated apparatus for 16 carrying out the process of vaporizing material into vaporized 17 particles and thereafter magnetizing the particles by subJecting 18 them to a magnetic field.
19 The chamber 155 is a vacuated chamber haying positioned in its lower half portion a pair of electrodes 160 and 162 of 21 magnetizable material. A source of voltage lS0 and 152 provides 22 voltage and current of opposite polarity via terminals 154 and 23 156 to the electrodes 160 and 162 via wire connections 164 and 24 166~ The area 161 between the end of the electrodes 160 and 162 is the spark gap.

~ ` lZ13671 l Upon the application of power to the magnetizable material 2 electrodes 160 and 162 the tip of the electrodes in the spark 3 gap will be vaporized into particles 180.
4 The particles 180 rise, use and enter into non-magnetic pipe 165. As the particles progress in movement they pass 6 between the magnetic field generator 175. The particles each 7 take on a magnetic charge as magnetized particles 185 and there-f~ 8 after are discharged via port 190 to the electrical particle 9 generator above described.
In the simplified preferred embodiment of Figures 1 and 2, 11 as well as the referred preferred embodiments it was indicated 12 that a low voltage was applied to the particle accelerator lO.
13 Upon acceleration a high voltage/current would be induced in the 14 secondary pickup coil 20. A most significant advantage of the present invention is that the voltage amplification is irrespec-16 tive to the wave-shape of the input voltage. Specifically, if 17 the input voltage is direct current a direct current voltage 18 will be at the output; an aLternating voltage will result in 19 an alternating voltage at the output; a pulsed voltage will result in a pulsed voltage at the output; and a voltage of any 21 other configuration will result in a similar configuration at 22 the output.
23 Although certain embodiments have been shown it is to be 24 understood modifications may be had without departing from the spirit and scope of the invention.

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Patent Citations (5)

Publication number	Priority date	Publication date	Assignee	Title
Family To Family Citations				
DE544908C *	1927-11-15	1932-02-23	J P Arend	Process for generating electrical currents
US3034002A *	1958-06-17	1962-05-08	Honeywell Regulator Co	Conductive fluid power transformer
US3300663A *	1963-12-30	1967-01-24	Nils O Rosaen	Power transfer circuits
US3859789A *	1972-01-31	1975-01-14	Battelle Development Corp	Method and apparatus for converting one form of energy into another form of energy
US4064409A *	1976-07-28	1977-12-20	Redman Charles M	Ferrofluidic electrical generator

* Cited by examiner, † Cited by third party

Cited By (2)

Publication number	Priority date	Publication date	Assignee	Title
Family To Family Citations				
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BR102012000838A2 *	2012-01-13	2015-03-31	Evoluç Es En Ltda	Earth's electron pickup

* Cited by examiner, † Cited by third party, ‡ Family to family citation

Similar Documents

Publication	Publication Date	Title
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US4613304A	1986-09-23	Gas electrical hydrogen generator
JPH08505278A	1996-06-04	Magnetic bearing system and method
CN1044942C	1999-09-01	Method and apparatus for forming magnetized zones on magnetizable body
US20230369954A1	2023-11-16	Apparatus, systems, and methods for generating force in electromagnetic system
US4577156A	1986-03-18	Push-pull betatron pair
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US4914412A	1990-04-03	Magnetic circuit

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CA1213671A	1986-11-04	Electrical particle generator
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SU693965A1	1982-08-30	Accelerator tube
SU746964A1	1980-07-23	Betatron magnetic system
JPS57166869A	1982-10-14	Motor with frequency generator

Priority And Related Applications

Applications Claiming Priority (2)

Application	Filing date	Title
US36705182A	1982-04-09	
US367,051	1982-04-09	

Legal Events

Date	Code	Title	Description
2003-11-04	MKEX	Expiry	

Concepts

machine-extracted

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Name	Image	Sections	Count	Query match
particle		title,claims,abstract,description	62	0.000
winding		claims,abstract,description	45	0.000
inductive effect		claims,abstract,description	10	0.000
magnetic particle		claims,description	5	0.000
material		claims,description	5	0.000
solid		claims,description	3	0.000
fluid		claims,description	2	0.000
liquid		claims,description	2	0.000
rubber tapping		claims	1	0.000
increasing effect		abstract,description	5	0.000

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